Unit 6 Notes

Right Triangles & Trigonometry

How to use this file: Copy the notes carefully, then cover the worked examples and redo them without looking. For practice files, work first, then check the answer bank. Detailed solutions are included for Unit 3 through Unit 8 practice as requested.

Aligned to the Honors Geometry unit outline provided by the user.

Unit 6 Notes: Right Triangles and Trigonometry

Big goal

Right-triangle geometry connects similarity, the Pythagorean Theorem, special side ratios, and trigonometric functions. Trigonometry turns side ratios into tools for finding missing sides and angles.

1. Pythagorean Theorem and converse

In a right triangle with legs a and b and hypotenuse c:

$$a^2 + b^2 = c^2.$$

The converse says that if three side lengths satisfy $a^2 + b^2 = c^2$ with c largest, then the triangle is right.

Common triples include $(3, 4, 5)$, $(5, 12, 13)$, $(8, 15, 17)$, $(7, 24, 25)$, and multiples of these.

2. Special right triangles

Triangle	Side relationship
$45° - 45° - 90°$	Legs are equal. If each leg is x, hypotenuse is $x\sqrt{2}$. If hypotenuse is h, each leg is $h/\sqrt{2} = \frac{h\sqrt{2}}{2}$.
$30° - 60° - 90°$	Short leg opposite $30°$ is x, hypotenuse is $2x$, and long leg opposite $60°$ is $x\sqrt{3}$.

3. Trigonometric ratios

In a right triangle, relative to an acute angle θ:

SOH-CAH-TOA

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

These ratios depend only on the angle, because all right triangles with the same acute angle are similar.

4. Inverse trig

Use inverse trig to find an angle:

$$\theta = \sin^{-1}\left(\frac{\text{opposite}}{\text{hypotenuse}}\right), \quad \theta = \cos^{-1}\left(\frac{\text{adjacent}}{\text{hypotenuse}}\right), \quad \theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right).$$

Check calculator mode: geometry angle problems usually use degrees.

5. Complementary angles

The acute angles in a right triangle are complementary. The sine of an angle equals the cosine of its complement:

$$\sin\theta = \cos(90° - \theta).$$

For example, $\sin 28° = \cos 62°$.

6. Applications

Angles of elevation look upward from a horizontal line. Angles of depression look downward from a horizontal line. In real-world right triangle problems, draw a diagram, label the angle, identify opposite/adjacent/hypotenuse, choose a trig ratio, and solve.

7. Area with included angle

For any triangle with sides a and b enclosing angle C,

$$A = \frac{1}{2}ab\sin C.$$

This comes from drawing an altitude: the height is $b\sin C$, so area is $\frac{1}{2}a(b\sin C)$.

8. Law of Sines and Law of Cosines

For any triangle with side a opposite angle A, side b opposite B, and side c opposite C:

Non-right triangle formulas

Law of Sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Law of Cosines:

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

Use Law of Sines when you know an angle-opposite side pair. Use Law of Cosines for SSS or SAS situations.

Ambiguous SSA case

SSA can sometimes produce zero, one, or two possible triangles. Always think carefully when using Law of Sines with two sides and a non-included angle.

9. Choosing a method

- Right triangle with two sides: Pythagorean Theorem.
- Right triangle with an acute angle and a side: trig ratio.
- Special angles $30°, 45°, 60°$: special right triangle ratios may give exact answers.
- Non-right triangle with ASA, AAS, or SSA: Law of Sines, with caution for SSA.
- Non-right triangle with SSS or SAS: Law of Cosines.

10. Worked examples

Example 1: Missing side

A right triangle has hypotenuse 13 and one leg 5. The other leg is

$$\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12.$$

Example 2: Trig side

If an angle is 35° and the hypotenuse is 20, the opposite side is

$$20 \sin 35^\circ \approx 11.5.$$

Example 3: Law of Cosines

With sides 7 and 9 and included angle 60°,

$$c^2 = 7^2 + 9^2 - 2(7)(9)\cos 60^\circ = 49 + 81 - 63 = 67.$$

Thus $c = \sqrt{67} \approx 8.2$.

Mastery checklist

You are ready for Unit 6 when you can solve right triangles, use exact special triangle ratios, apply inverse trig, solve applications, compute area with $\frac{1}{2}ab\sin C$, and choose between Law of Sines and Law of Cosines.